

Safe and Reliable ICs for EPS Systems



Allegro MicroSystems provides a full array of integrated circuits (ICs) for use in Electric Power Steering (EPS) systems. Allegro has a long history in the automotive industry (more than 4 billion ICs shipped) and provides cutting edge sensor and power IC solutions that meet the needs of advanced EPS systems. Allegro is a high volume supplier of Hall-effect sensor ICs for torque, steering wheel position, motor current, and motor position sensing applications. Allegro's power ICs are designed for safety and include advanced features to diagnose IC performance **to ensure that system safety is never compromised**. Allegro's complete line of power management ICs is designed specifically for automotive safety systems and complements Allegro's motor driver ICs and sensor ICs for a robust and reliable EPS system.

Features/Benefits:

- AEC-Q100 Grade 0 qualified.
- Power Management ICs provide multiple output voltages from wide V_{in} , with buck or buck-boost pre-regulator for battery voltages including idle-stop/load-dump conditions
- Angle sensor ICs integrate a steering wheel turns counter, and a micropower mode of operation.
- Sensor ICs can be operated from a 12 V battery.
- Sensors support a wide range of output formats: analog, SPI, I²C, SENT, high speed SENT.
- Diagnostic and redundant die options to support ASIL compliance.
- Gate driver ICs include redundant high-performance low side current sense amplifiers.
- High accuracy current sensor IC solutions support low or high side current sensing with near zero insertion loss.
- Advanced diagnostics and bridge control parameters can be programmed via industry standard serial interface.
- Supports systems using both DC brushed and brushless motor types.

- Development process is certified to the requirements of ISO 26262.



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